

RT-DETRv4: Painlessly Furthering Real-Time Object Detection with Vision Foundation Models

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Abstract

Real-time object detection has achieved substantial progress through meticulously designed architectures and optimization strategies. However, the pursuit of high-speed inference via lightweight network designs often leads to degraded feature representation, which hinders further performance improvements and practical on-device deployment. In this paper, we propose a cost-effective and highly adaptable distillation framework that harnesses the rapidly evolving capabilities of Vision Foundation Models (VFM) to enhance lightweight object detectors. Given the significant architectural and learning objective disparities between VFMs and resource-constrained detectors, achieving stable and task-aligned semantic transfer is challenging. To address this, on one hand, we introduce a **Deep Semantic Injector (DSI)** module that facilitates the integration of high-level representations from VFMs into the deep layers of the detector. On the other hand, we devise a **Gradient-guided Adaptive Modulation (GAM)** strategy, which dynamically adjusts the intensity of semantic transfer based on gradient norm ratios. Without increasing deployment and inference overhead, our approach painlessly delivers striking and consistent performance gains across diverse DETR-based models, underscoring its practical utility for real-time detection. Our new model family, RT-DETRv4, achieves state-of-the-art results on COCO, attaining AP scores of 49.7/53.5/55.4/57.0 at corresponding speeds of 273/169/124/78 FPS.

1. Introduction

Real-time object detection stands as a fundamental task in computer vision, which underpins numerous interactive and safety-critical applications that demand instant perception and decision making, such as autonomous driving [5], embodied intelligence [21], and human-computer interac-

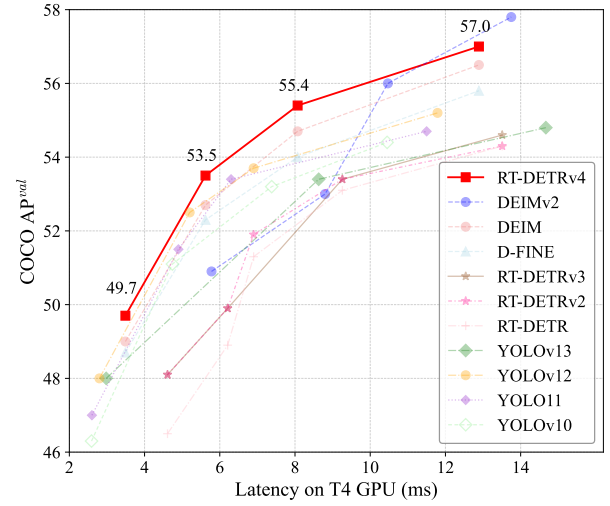


Figure 1. Compared with existing advanced real-time object detectors on COCO [20]. Our RT-DETRv4 models achieve state-of-the-art performance.

tion [16]. Over the past decade, remarkable progress has been driven by increasingly efficient network architectures and end-to-end learning frameworks. In particular, two representative series, YOLO [30] and DETR [3], have profoundly influenced the evolution of object detection paradigms. The YOLO family emphasizes rapid one-stage detection achieving high inference speed and practical deployment efficiency, and the DETR series has reshaped the detection paradigm through its unified modeling of object queries and set-based prediction. Among its variants, RT-DETR [38] marked a milestone as the first real-time DETR, introducing the DETR family to the real-time community by outperforming YOLO models in both speed and accuracy.

Despite the remarkable progress, a long-standing challenge remains: the inherent trade-off between designing lightweight models to achieve high inference speed and employing complex architectures to improve feature representa-

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RT-DETRv4：基于视觉基础模型轻松推进实时目标检测

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摘要

Real-time object detection has achieved substantial progress through meticulously designed architectures and optimization strategies. However, the pursuit of high-speed inference via lightweight network designs often leads to degraded feature representation, which hinders further performance improvements and practical on-device deployment. In this paper, we propose a cost-effective and highly adaptable distillation framework that harnesses the rapidly evolving capabilities of Vision Foundation Models (VFM) to enhance lightweight object detectors. Given the significant architectural and learning objective disparities between VFMs and resource-constrained detectors, achieving stable and task-aligned semantic transfer is challenging. To address this, on one hand, we introduce a **Deep Semantic Injector (DSI)** module that facilitates the integration of high-level representations from VFMs into the deep layers of the detector. On the other hand, we devise a **Gradient-guided Adaptive Modulation (GAM)** strategy, which dynamically adjusts the intensity of semantic transfer based on gradient norm ratios. Without increasing deployment and inference overhead, our approach painlessly delivers striking and consistent performance gains across diverse DETR-based models, underscoring its practical utility for real-time detection. Our new model family, RT-DETRv4, achieves state-of-the-art results on COCO, attaining AP scores of 49.7/53.5/55.4/57.0 at corresponding speeds of 273/169/124/78 FPS.

1. 引言

实时目标检测是计算机视觉中的一项基础任务，它支撑着众多需要即时感知与决策的交互式和安全关键型应用，例如自动驾驶[5]、具身智能[21]以及人机交互—

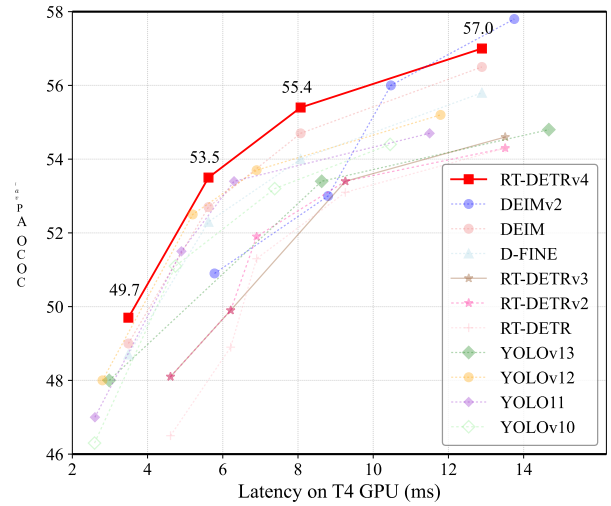


图1. 与COCO数据集[20]上现有先进实时目标检测器相比，我们的RT-DETRv4模型实现了最先进的性能。

tion [16]。过去十年中，网络架构的日益高效和端到端学习框架推动了显著进展。特别是YOLO [30]和DETR [3]这两个代表性系列，深刻影响了目标检测范式的演进。YOLO系列强调快速单阶段检测，实现了高推理速度和实际部署效率；而DETR系列则通过对目标查询的统一建模和基于集合的预测，重塑了检测范式。在其变体中，RT-DETR [38]作为首个实时DETR模型具有里程碑意义，它以速度和精度均超越YOLO模型的表现，将DETR系列引入了实时检测领域。

尽管取得了显著进展，一个长期存在的挑战依然存在：设计轻量级模型以实现高推理速度与采用复杂架构以提升特征表示能力之间的固有权衡。

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tion. To meet real-time constraints, detectors typically adopt lightweight backbones and carefully designed computational modules, which inevitably reduce their ability to capture high-level semantics and lead to a semantic bottleneck. This limitation not only hinders further performance improvement but also increases the difficulty of practical on-device deployment.

In this paper, inspired by the rapid advances in Vision Foundation Models (VFMs) [13, 31], we propose a cost-effective and highly adaptable distillation framework that leverages the powerful representational capacity of VFMs to enhance lightweight object detectors. By transferring the rich semantics of VFMs to real-time detectors during training while keeping the detector architecture unchanged during inference, our method enables significant enhancement without introducing any additional inference or deployment cost. This advantage is particularly important for practical real-time detection applications.

However, achieving stable and task-aligned semantic transfer is challenging because of the large architectural and learning objective disparities between VFMs and resource-constrained detectors. To address this issue, we first introduce a Deep Semantic Injector (DSI) module that enables the integration of high-level representations from VFMs into the deep layers of the detector. To ensure stable and efficient optimization, we further design a Gradient-guided Adaptive Modulation (GAM) strategy that dynamically adjusts the strength of semantic injection based on gradient norm ratios, thereby harmonizing the learning of semantic transfer and detection objectives.

Extensive experiments demonstrate that the proposed framework achieves consistent and significant performance improvements over advanced DETR-based detectors without increasing inference or deployment overhead, underscoring its effectiveness. In summary, our main contributions are as follows:

- We propose a cost-effective and highly adaptable distillation framework that leverages the evolving capabilities of VFMs to painlessly enhance real-time detectors, providing a scalable pathway for transferring foundation-level semantics to lightweight architectures.
- We propose the Deep Semantic Injector (DSI) and Gradient-guided Adaptive Modulation (GAM), which enable stable and task-aligned semantic transfer between VFMs and detectors with significantly different architectures and learning objectives.
- We establish a new family of models, RT-DETRv4-S/M/L/X, achieving 49.7/53.5/55.4/57.0 AP scores on COCO [20] at 273/169/124/78 FPS, setting a new SOTA on COCO dataset.

2. Related Work

2.1. Real-time Object Detection

The evolution of real-time object detection has long been driven by the You Only Look Once (YOLO) family [30], which popularized the single-stage paradigm through an efficient and unified detection pipeline. Over the past few years, this lineage has undergone rapid iteration, introducing continuous refinements in backbone design, label assignment, and optimization strategy [2, 9, 10, 18, 28, 29, 34, 35]. Recent generations have expanded the design space even further: YOLOv10 [33] eliminated NMS that the YOLO series has long relied on, YOLO11 [11] improved the architectural hierarchy and neck connectivity, YOLOv12 [32] incorporated attention mechanisms for better contextual reasoning, and YOLOv13 [17] explored hypergraph representations to capture higher-order feature dependencies. These advances have pushed the performance–efficiency frontier of convolutional and hybrid architectures, gradually narrowing the gap between real-time and high-accuracy detectors.

In parallel, another line of research has evolved around the DEtection TRansformer (DETR) [3], which redefined object detection as a set prediction problem and eliminated hand-crafted components such as anchor design and NMS. This transformer-based paradigm inspired numerous variants, including Deformable DETR [39], Conditional DETR [24], and DAB-DETR [22], which focus on improving convergence and localization accuracy. Later works such as DN-DETR [19], DINO [37], and Group-DETR [6] introduced denoising objectives and group-wise supervision to further enhance training stability and representational quality.

Building on this foundation, RT-DETR [38] established the first real-time end-to-end transformer detector that achieved parity with, and in some cases surpassed, contemporary YOLO models. Subsequent works have continued to improve its training efficiency and representation learning without incurring inference overhead. For instance, RT-DETRv2 [23] and RT-DETRv3 [36] incorporated auxiliary supervision for enhanced gradient flow, D-FINE [26] employed self-distillation to refine semantic representation, and DEIM [15] introduced dense matching for more precise feature alignment. Collectively, these developments illustrate a clear trend: as architectural efficiency saturates, training supervision and semantic representation become the primary levers for further progress. Our work builds on this insight by strengthening the core representation via deep semantic transfer, achieving higher accuracy at no additional deployment or inference cost.

2.2. Vision Foundation Models

Vision Foundation Models (VFMs) have become a dominant paradigm for learning general-purpose vision representations from large-scale image corpora with minimal or no

为了满足实时性约束，检测器通常采用轻量级骨干网络和精心设计的计算模块，这不可避免地降低了其捕获高层语义的能力，并导致语义瓶颈。这一限制不仅阻碍了性能的进一步提升，也增加了实际设备端部署的难度。

本文受视觉基础模型（VFM）[13, 31]快速发展的启发，提出了一种成本效益高、适应性强的蒸馏框架，利用VFM强大的表征能力来增强轻量级目标检测器。通过在训练阶段将VFM丰富的语义信息迁移到实时检测器中，同时在推理阶段保持检测器架构不变，我们的方法能够在不引入任何额外推理或部署成本的情况下实现显著性能提升。这一优势对于实际实时检测应用尤为重要。

然而，由于视觉基础模型与资源受限检测器之间在架构和学习目标上存在巨大差异，实现稳定且与任务对齐的语义迁移具有挑战性。为解决这一问题，我们首先引入了深度语义注入器模块，该模块能够将视觉基础模型的高层表征整合到检测器的深层网络中。为确保稳定高效的优化，我们进一步设计了梯度引导自适应调制策略，该策略基于梯度范数比动态调整语义注入的强度，从而协调语义迁移与检测目标的学习过程。

大量实验表明，所提出的框架在不增加推理或部署开销的情况下，相比先进的基于DETR的检测器实现了持续且显著的性能提升，凸显了其有效性。总而言之，我们的主要贡献如下：

- 我们提出了一种成本效益高且高度自适应的蒸馏框架，该框架利用视觉基础模型不断演进的能力，轻松增强实时检测器，为将基础级语义迁移至轻量级架构提供了可扩展的路径。
- 我们提出了深度语义注入器（DSI）和梯度引导自适应调制（GAM），它们能够在架构和学习目标显著不同的视觉基础模型（VFM）与检测器之间实现稳定且任务对齐的语义迁移。
- 我们建立了一个新的模型系列RT-DETRv4-S/M/L/X，在COCO数据集上以273/169/124/78 FPS的速度实现了49.7/53.5/55.4/57.0 AP的得分，为COCO数据集设立了新的SOTA。

2. 相关工作

2.1. 实时目标检测

实时目标检测的发展长期以来一直由YOLO（You Only Look Once）系列[30]推动，该系列通过高效统一的检测流程普及了单阶段检测范式。过去几年间，该系列经历了快速迭代，在骨干网络设计、标签分配和优化策略等方面持续改进[2, 9, 10, 18, 28, 29, 34, 35]。最新版本进一步拓展了设计空间：YOLOv10[33]摒弃了YOLO系列长期依赖的非极大值抑制，YOLO11[11]优化了架构层次与颈部连接，YOLOv12[32]引入注意力机制以增强上下文推理能力，YOLOv13[17]则探索超图表示来捕获高阶特征依赖。这些进展推动了卷积与混合架构在性能与效率边界上的突破，逐步缩小了实时检测器与高精度检测器之间的差距。

与此同时，另一条研究路线围绕DEtection TRansformer（DETR）[3]展开，它将目标检测重新定义为集合预测问题，并消除了锚点设计和非极大值抑制等手工设计组件。这种基于Transformer的范式催生了众多变体，包括Deformable DETR [39]、Conditional DETR [24]和ID AB-DETR [22]，这些方法专注于提升收敛速度与定位精度。后续研究如DN-DETR [19]、DINO [37]和Group-DETR [6]则引入了去噪目标和分组监督机制，进一步增强了训练稳定性和表征质量。

在此基础上，RT-DETR [38] 构建了首个实时端到端Transformer检测器，其性能与同期YOLO模型持平，并在某些情况下实现了超越。后续研究持续提升了其训练效率和表征学习能力，且未增加推理开销。例如，RT-DETRv2 [23] 和 RT-DETRv3 [36] 通过辅助监督增强梯度流，D-FINE [26] 采用自蒸馏技术优化语义表征，而DEIM [15] 则引入密集匹配以实现更精确的特征对齐。这些进展共同揭示了一个明显趋势：当架构效率趋于饱和时，训练监督与语义表征成为推动性能提升的主要杠杆。本研究基于这一洞见，通过深度语义迁移强化核心表征，在不增加部署或推理成本的前提下实现了更高的精度。

2.2. 视觉基础模型

视觉基础模型（VFM）已成为从大规模图像语料库中以最少或无需监督的方式学习通用视觉表示的主导范式。

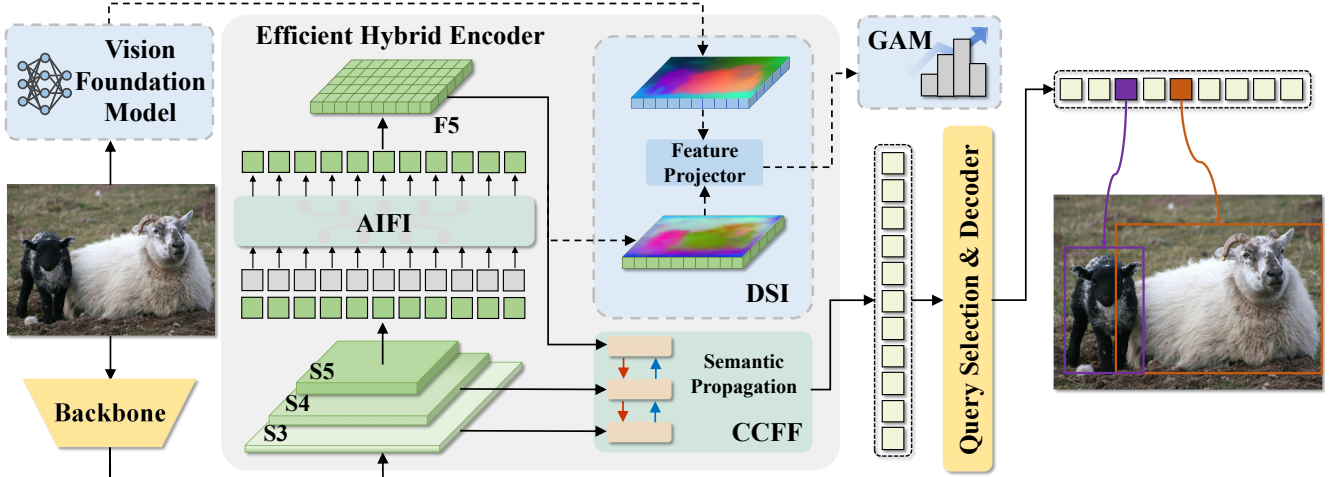


Figure 2. **Overview of RT-DETRv4.** We leverage a Vision Foundation Model (VFM) to extract high-quality semantic representations, which are aligned with the deepest feature map (F_5) from the AIFI module via a Feature Projector in the Deep Semantic Injector (DSI). To ensure faster and more stable convergence, a Gradient-guided Adaptive Modulation (GAM) dynamically adjusts the DSI loss during training. The proposed framework operates only during the training phase (highlighted by dashed arrows and blue blocks) of the real-time detector and keeps the original architecture unchanged during inference and deployment, introducing no additional overhead while improving accuracy.

human supervision. Early progress stemmed from self- and weakly-supervised learning methods, which enabled models to capture high-level semantics from unlabeled or loosely labeled data. Representative approaches include contrastive learning frameworks such as SimCLR [7] and MoCo [12], which learn discriminative features by enforcing consistency across augmented views of the same image while contrasting them with others. CLIP [27] further extended this idea to large-scale image–text contrastive training, aligning visual and linguistic embeddings and demonstrating strong zero-shot transferability across diverse tasks.

Inspired by masked language modeling in NLP, Masked Image Modeling (MIM) approaches were introduced to reconstruct masked image regions, thereby learning context-aware and holistic representations. Notable methods include MAE [13] and BEiT [1]. Building on these advances, the DINO family [4, 25, 31] integrates contrastive, reconstruction-based, and self-distillation objectives to produce highly semantic and transferable features. In particular, DINOv3 [31] demonstrates the scalability and efficacy of large-scale self-supervised learning, achieving rich and robust representations without human annotations.

3. Method

3.1. Overview

In this work, we focus on applying our framework to DETR-based real-time object detectors, *i.e.*, RT-DETR models [38]. The overall framework of our method is shown in Figure 2, where the proposed modules are highlighted in blue.

Preliminaries. Our model builds upon the RT-DETR architecture, particularly its efficient hybrid encoder, as illustrated in the overall framework. The encoder processes multi-scale feature maps (S_3, S_4, S_5) extracted from a CNN backbone. It consists of two main components:

- **Attention-based Intra-scale Feature Interaction (AIFI):** To maintain computational efficiency, self-attention is applied only to the highest-level feature map, $S_5 \in \mathbb{R}^{H/32 \times W/32 \times C_5}$. AIFI captures global context and long-range dependencies, producing an enhanced representation denoted as F_5 .
- **CNN-based Cross-scale Feature Fusion (CCFF):** The semantically enriched F_5 is further fused with the lower-level feature maps S_3 and S_4 to propagate high-level semantics to shallower features, generating the final multi-scale outputs P_3, P_4, P_5 for the decoder.

Motivation. The design of the hybrid encoder makes the quality of the feature map F_5 particularly critical. As the only feature subjected to self-attention, F_5 serves as the principal source of high-level, global semantic information for the entire model. Its quality directly affects the subsequent cross-scale fusion in the CCFF module, the initial query selection, and ultimately the performance of the decoder. This dependency leads to what we term the *F5 Semantic Bottleneck*. However, the AIFI module that produces F_5 is trained with only indirect supervision, as the gradients from the final detection losses must backpropagate through the decoder and CCFF before reaching F_5 . Such indirect supervision may be insufficient to fully optimize F_5 .

To address this issue, we propose the **Deep Semantic Injector (DSI)**, a lightweight training-only module that ex-

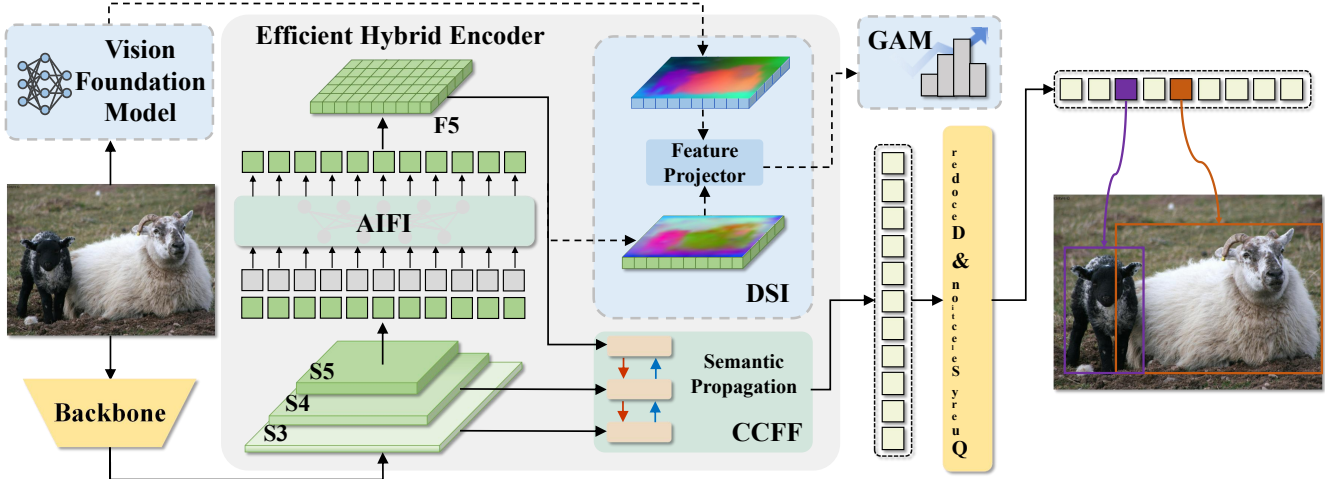


图2. RT-DETRv4概览。我们利用视觉基础模型（VFM）提取高质量的语义表示，这些表示通过深度语义注入器（DSI）中的特征投影器与AIFI模块的最深层特征图（ F_5 ）对齐。为确保更快且更稳定的收敛，梯度引导自适应调制（GAM）在训练过程中动态调整DSI损失。所提出的框架仅在实时检测器的训练阶段（由虚线箭头和蓝色块突出显示）运行，并在推理和部署期间保持原始架构不变，在提升精度的同时不引入额外开销。

人类监督。早期的进展源于自监督和弱监督学习方法，这些方法使模型能够从未标记或松散标记的数据中捕获高级语义。代表性方法包括对比学习框架，如SimCLR [7]和MoCo [12]，它们通过增强同一图像的不同视图之间的一致性，同时与其他图像形成对比，来学习判别性特征。CLIP [27]进一步将这一思想扩展到大规模的图像-文本对比训练中，对齐视觉和语言嵌入，并在多样化的任务中展现出强大的零样本迁移能力。

受自然语言处理中掩码语言建模的启发，掩码图像建模方法被引入以重建被遮蔽的图像区域，从而学习上下文感知和整体性表征。其中值得注意的方法包括MAE [13]和BEiT [1]。在这些进展的基础上，DINO系列方法[4, 25, 31]融合了对比学习、基于重建的目标以及自蒸馏目标，以产生高度语义化且可迁移的特征。特别是DINOv3 [31]证明了大规模自监督学习的可扩展性和有效性，无需人工标注即可获得丰富且鲁棒的表征。

3. 方法

3.1. 概述

在这项工作中，我们专注于将我们的框架应用于基于DETR的实时目标检测器，*i.e.*，即RT-DETR模型[38]。我们方法的整体框架如图2所示，其中所提出的模块以蓝色高亮显示。

预备知识。我们的模型基于RT-DETR架构，特别是其高效的混合编码器，如整体框架所示。编码器处理从CNN骨干网络提取的多尺度特征图（ S_3 、 S_4 、 S_5 ）。它包含两个主要组成部分：

- 基于注意力的尺度内特征交互（AIFI）：为保持计算效率，自注意力机制仅应用于最高层级的特征图 $S_5 \in \mathbb{R}^{H/32 \times W/32 \times C_5}$ 。AIFI能够捕获全局上下文和长程依赖关系，生成一个增强的表示，记为 F_5 。
- 基于CNN的跨尺度特征融合（CCFF）：语义增强的 F_5 进一步与低层特征图 S_3 和 S_4 融合，将高层语义传播至浅层特征，最终生成解码器所需的多尺度输出 P_3, P_4, P_5 。

动机。混合编码器的设计使得特征图 F_5 的质量尤为关键。作为唯一经过自注意力机制处理的特征， F_5 是整个模型获取高层次、全局语义信息的主要来源。其质量直接影响后续CCFF模块中的跨尺度融合、初始查询选择，并最终决定解码器的性能。这种依赖性导致了我们的 F_5 *Semantic Bottleneck* 问题。然而，生成 F_5 的AIFI模块仅通过间接监督进行训练，因为最终检测损失的梯度需经过解码器和CCFF模块反向传播才能到达 F_5 。这种间接监督可能不足以充分优化 F_5 。

为了解决这一问题，我们提出了深度语义注入器（DSI），这是一个轻量级的仅训练模块，旨在

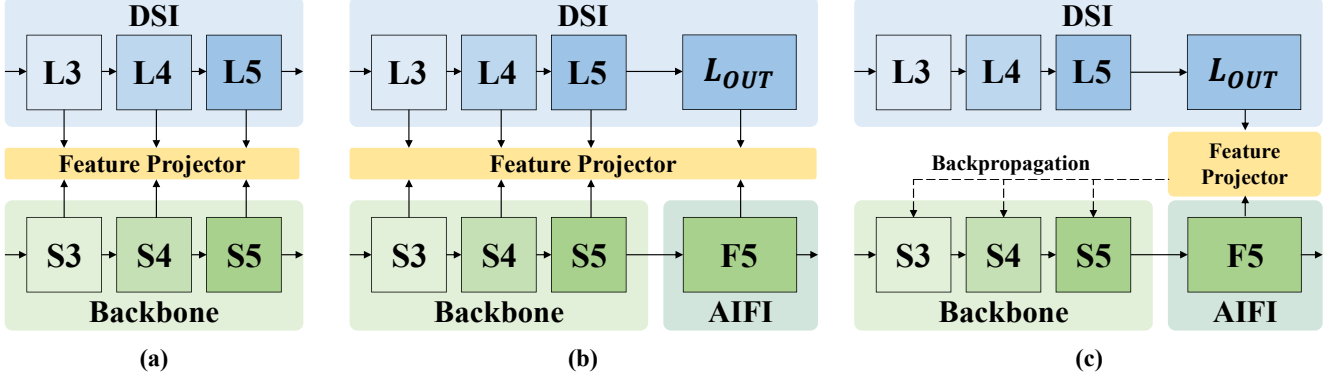


Figure 3. **Illustration of different Deep Semantic Injector (DSI) strategies.** (a) Direct alignment of multi-scale backbone features (S_3, S_4, S_5). (b) Hybrid alignment of both backbone features and the AIFI output feature (F_5). (c) Our proposed method: Targeted alignment of only the AIFI output feature (F_5), which possesses the highest-level semantics. This design allows gradients to backpropagate, enhancing both the AIFI module and the backbone.

explicitly aligns the deep feature F_5 with semantically rich representations from a vision foundation model. This targeted supervision enhances the semantic expressiveness of F_5 and allows its gradient to flow back through AIFI and the backbone, improving both modules synergistically.

With DSI incorporated, the total training objective is defined as:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{det}} + \lambda \mathcal{L}_{\text{DSI}}, \quad (1)$$

where $\mathcal{L}_{\text{det}}$ denotes the standard detection loss (e.g., classification and bounding box regression), and $\mathcal{L}_{\text{DSI}}$ represents the proposed semantic alignment loss. However, achieving stable and task-aligned semantic transfer is challenging because of the large architectural and learning objective disparities between VFMs and resource-constrained detectors. An inappropriate choice of λ may either provide insufficient semantic supervision in the early stages or excessively dominate the detection objective in later stages, ultimately impeding convergence and degrading performance. To adapt to the evolving optimization dynamics during training, we propose **Gradient-guided Adaptive Modulation (GAM)**, a mechanism that dynamically tunes λ based on gradient statistics, ensuring balanced optimization between detection and semantic supervision.

3.2. Deep Semantic Injector

To address the *F5 Semantic Bottleneck*, we introduce the Deep Semantic Injector (DSI), a training-only module designed to provide explicit and powerful supervision for the feature map F_5 . The objective of DSI is to enrich the semantic quality of F_5 by aligning it with representations from a high-capacity semantic teacher, denoted as $\mathcal{T}$. Given an input image, $\mathcal{T}$ produces a high-quality feature representation $F_{\mathcal{T}} \in \mathbb{R}^{H' \times W' \times C_{\mathcal{T}}}$.

Feature Projector. To align the detector’s feature map $F_5 \in \mathbb{R}^{H_5 \times W_5 \times C_5}$ with the teacher’s representation $F_{\mathcal{T}}$,

differences in both spatial resolution and channel dimensionality must be reconciled. The teacher, typically a ViT [8], outputs a sequence of patch tokens $T_p \in \mathbb{R}^{N_p \times C_{\mathcal{T}}}$. To enable spatial comparison, T_p is reshaped into a 2D grid representation $F_{\mathcal{T}}^{\text{sp}} \in \mathbb{R}^{H_{\mathcal{T}} \times W_{\mathcal{T}} \times C_{\mathcal{T}}}$, where $N_p = H_{\mathcal{T}} \times W_{\mathcal{T}}$.

We then introduce a lightweight feature projector $\mathcal{P}$ to achieve twofold alignment. First, $F_{\mathcal{T}}^{\text{sp}}$ is interpolated to match the spatial resolution of F_5 . Meanwhile, $\mathcal{P}$ adjusts the channel dimension of F_5 to align with the teacher’s semantic space. The complete projection process is summarized as follows:

$$F_{\mathcal{T}}^{\text{sp}} = \text{Reshape}(T_p), \quad F_{\mathcal{T}}^{\text{sp}} \in \mathbb{R}^{H_{\mathcal{T}} \times W_{\mathcal{T}} \times C_{\mathcal{T}}}, \quad (2)$$

$$F'_{\mathcal{T}} = \text{Interpolate}(F_{\mathcal{T}}^{\text{sp}}), \quad F'_{\mathcal{T}} \in \mathbb{R}^{H_5 \times W_5 \times C_{\mathcal{T}}}, \quad (3)$$

$$F'_5 = \mathcal{P}(F_5), \quad F'_5 \in \mathbb{R}^{H_5 \times W_5 \times C_{\mathcal{T}}} \quad (4)$$

Semantic Injection. As illustrated in Figure 3, we design three progressively enhanced configurations for semantic injection. In configurations (a) and (b), the DSI module performs feature alignment at different hierarchical depths through the Feature Projector, injecting semantic knowledge from the frozen VFM into the detector’s feature hierarchy. In configuration (c), considering the pivotal role of the AIFI module within the hybrid encoder, the alignment is conducted on its output F_5 , which contains the richest semantics. Without detaching gradients, the DSI loss is allowed to propagate backward, thereby updating the lightweight backbone. Consequently, the forward pass leverages the enriched F_5 to guide cross-scale fusion in the CCFF module, while the backward path enforces semantic consistency and strengthens the backbone’s representational capacity. This dual-directional supervision achieves a unified semantic enhancement of the detector.

Alignment Loss. To encourage the detector to capture the rich semantic of the teacher’s representation, we adopt a

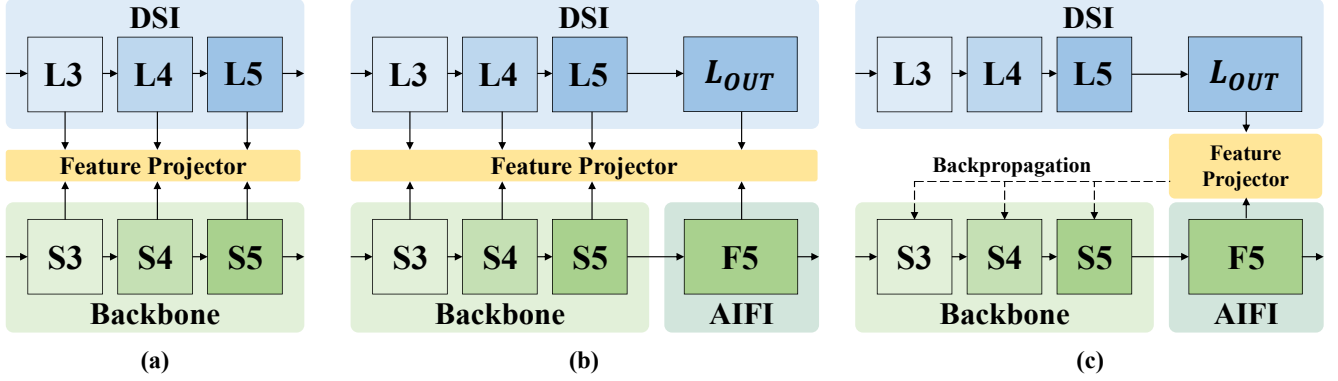


图3. 不同深度语义注入器 (DSI) 策略示意图。 (a) 多尺度骨干特征直接对齐 (S_3, S_4, S_5)。 (b) 骨干特征与AIFI输出特征的混合对齐 (F_5)。 (c) 我们提出的方法：仅针对具有最高层级语义的AIFI输出特征进行定向对齐 (F_5)。该设计允许梯度反向传播，从而同时增强AIFI模块与骨干网络。

explicitly aligns the deep feature F_5 with semantically rich representations from a vision foundation model. This targeted supervision enhances the semantic expressiveness of F_5 and allows its gradient to flow back through AIFI and the backbone, improving both modules synergistically.

在融入DSI后，总训练目标被定义为：

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{det}} + \lambda \mathcal{L}_{\text{DSI}}, \quad (1)$$

其中 $\mathcal{L}_{\text{det}}$ 表示标准检测损失（如分类和边界框回归）， $\mathcal{L}_{\text{DSI}}$ 代表提出的语义对齐损失。然而，由于视觉基础模型与资源受限检测器之间巨大的架构和学习目标差异，实现稳定且任务对齐的语义传递具有挑战性。 λ 的选择不当可能导致早期阶段语义监督不足，或在后期阶段过度主导检测目标，最终阻碍收敛并降低性能。为适应训练过程中不断变化的优化动态，我们提出梯度自适应调制机制，该机制基于梯度统计动态调整 λ ，确保检测目标与语义监督之间的平衡优化。

3.2. 深度语义注入器

为了解决 F_5 *Semantic Bottleneck*，我们引入了深度语义注入器 (DSI)，这是一个仅用于训练的模块，旨在为特征图 F_5 提供显式且强大的监督。DSI的目标是通过将 F_5 与来自高性能语义教师（表示为 $\mathcal{T}$ ）的表征对齐，来丰富 F_5 的语义质量。给定输入图像， $\mathcal{T}$ 会生成高质量的特征表示 $F_{\mathcal{T}} \in \mathbb{R}^{H' \times W' \times C_{\mathcal{T}}}$ 。

特征投影器。为了将检测器的特征图 $F_5 \in \mathbb{R}^{H_5 \times W_5 \times C_5}$ 与教师的表征 $F_{\mathcal{T}}$ 对齐，

空间分辨率和通道维度上的差异必须得到调和。教师模型（通常为ViT [8]）输出一系列图像块标记 $T_p \in \mathbb{R}^{N_p \times C_{\mathcal{T}}}$ 。为实现空间对比， T_p 被重塑为二维网格表示 $F_{\mathcal{T}}^{\text{sp}} \in \mathbb{R}^{H_{\mathcal{T}} \times W_{\mathcal{T}} \times C_{\mathcal{T}}}$ ，其中 $N_p = H_{\mathcal{T}} \times W_{\mathcal{T}}$ 。

接着，我们引入了一个轻量级特征投影器 $\mathcal{P}$ 来实现双重对齐。首先， $F_{\mathcal{T}}^{\text{sp}}$ 通过插值调整以匹配 F_5 的空间分辨率。同时， $\mathcal{P}$ 调整 F_5 的通道维度，使其与教师的语义空间对齐。完整的投影过程总结如下：

$$F_{\mathcal{T}}^{\text{sp}} = \text{Reshape}(T_p), \quad F_{\mathcal{T}}^{\text{sp}} \in \mathbb{R}^{H_{\mathcal{T}} \times W_{\mathcal{T}} \times C_{\mathcal{T}}}, \quad (2)$$

$$F'_{\mathcal{T}} = \text{Interpolate}(F_{\mathcal{T}}^{\text{sp}}), \quad F'_{\mathcal{T}} \in \mathbb{R}^{H_5 \times W_5 \times C_{\mathcal{T}}}, \quad (3)$$

$$F'_5 = \mathcal{P}(F_5), \quad F'_5 \in \mathbb{R}^{H_5 \times W_5 \times C_{\mathcal{T}}} \quad (4)$$

语义注入。如图3所示，我们设计了三种逐步增强的语义注入配置。在配置 (a) 和 (b) 中，DSI模块通过特征投影器在不同层级深度执行特征对齐，将来自冻结VFM的语义知识注入检测器的特征层次结构。在配置 (c) 中，考虑到AIFI模块在混合编码器中的关键作用，对齐操作在其输出 F_5 上进行，该输出包含最丰富的语义信息。在保持梯度连接的情况下，DSI损失得以反向传播，从而更新轻量级骨干网络。因此，前向传播利用增强后的 F_5 来指导CCFF模块中的跨尺度融合，而反向路径则强制语义一致性并增强骨干网络的表征能力。这种双向监督实现了检测器语义增强的统一。

对齐损失。为了鼓励检测器捕捉教师表示的丰富语义，我们采用了一种

cosine similarity loss. We maximize the patch-wise cosine similarity between the detector’s projected features F'_5 and the teacher’s projected features F'_T , formulated as minimizing the negative cosine similarity averaged over all spatial locations (i, j) :

$$\mathcal{L}_{DSI}(F'_5, F'_T) = -\frac{1}{H_5 W_5} \sum_{i,j} \frac{F'_5(i, j) \cdot F'_T(i, j)}{\|F'_5(i, j)\| \|F'_T(i, j)\|}. \quad (5)$$

3.3. Gradient-guided Adaptive Modulation

To ensure stable and adaptive semantic supervision, we propose a dynamic Gradient-guided Adaptive Modulation (GAM) mechanism that regulates the relative contribution of the AIFI module according to its *gradient norm ratio* rather than the raw loss magnitude. This gradient-based regulation adaptively maintains the effective contribution of AIFI within a desired range, leading to balanced optimization among model components.

Specifically, for each training step t within epoch e , we compute the L_1 norm of gradients for each major component, including the backbone, AIFI, CCFF, and decoder:

$$\mathcal{C} = \{\text{Backbone, AIFI, CCFF, Decoder}\}, \quad (6)$$

$$G_t^{(C)} = \|\nabla_{\theta_C} \mathcal{L}_{total}\|_1. \quad (7)$$

The total gradient magnitude is given by:

$$G_t^{(total)} = \sum_C G_t^{(C)}, \quad (8)$$

and the relative gradient contribution of AIFI at step t is defined as:

$$r_t = \frac{G_t^{(\text{AIFI})}}{G_t^{(total)}}. \quad (9)$$

We then average the gradient ratios across all training steps within epoch e to obtain:

$$\bar{r}_e = \frac{1}{T_e} \sum_{t=1}^{T_e} r_t, \quad (10)$$

where T_e denotes the number of steps in epoch e .

Two hyperparameters govern the modulation process:

- **Target Ratio** (ρ): the desired average gradient ratio of AIFI, representing its ideal relative contribution to optimization.
- **Tolerance Interval** (δ): a margin that defines an acceptable deviation range $[\rho - \delta, \rho + \delta]$ around the target ratio.

At the end of each epoch, GAM checks whether $\bar{r}_e$ lies within the target interval. If $\bar{r}_e \in [\rho - \delta, \rho + \delta]$, the weight λ_e of $\mathcal{L}_{DSI}$ remains unchanged. Otherwise, λ_e is adjusted such that the next epoch’s AIFI gradient ratio is steered toward the

further boundary of the target range rather than its midpoint, since only a portion of AIFI’s gradients originates from $\mathcal{L}_{DSI}$, boundary-based adjustment yields more stable convergence near equilibrium.

$$\lambda_{e+1} = \begin{cases} \lambda_e \cdot \frac{\rho - \delta}{\bar{r}_e}, & \text{if } \bar{r}_e > \rho + \delta, \\ \lambda_e \cdot \frac{\rho + \delta}{\bar{r}_e}, & \text{if } \bar{r}_e < \rho - \delta, \\ \lambda_e, & \text{otherwise.} \end{cases} \quad (11)$$

This update rule drives the effective gradient contribution of AIFI to converge within the desired operational range while preventing oscillations. The hyperparameters ρ and δ offer explicit control over training dynamics: ρ defines the desired supervision intensity, whereas δ regulates the trade-off between responsiveness and stability. A smaller δ enables faster adaptation but risks instability, while a larger δ yields smoother yet slower convergence. In practice, GAM provides stable convergence and consistently improves semantic alignment without additional tuning overhead.

4. Experiments

4.1. Setup

Dataset and Metric. All experiments are conducted on the COCO 2017 [20] dataset, using the `train2017` split for training and `val2017` for evaluation. We report the standard COCO metrics, including AP (averaged over uniformly sampled IoU thresholds ranging from 0.50-0.95 with a step size of 0.05), AP_{50} , AP_{75} , as well as AP at different scales: AP_S , AP_M , AP_L .

Implementation Details. Our experiments are based on the RT-DETR architecture [38], with additional architectural and training refinements from RT-DETRv2 [23], D-FINE [26], and DEIM [15]. For fair comparison, the core hyperparameters remain consistent with those in the corresponding baselines. The DSI employs a pre-trained and frozen DINOv3-ViT-B model as the semantic teacher. All evaluations are conducted using the COCO AP metrics, and inference latency (in milliseconds) is measured on a single NVIDIA T4 GPU under TensorRT FP16 precision.

4.2. Comparison with SOTA

We compare our proposed RT-DETRv4 with recent state-of-the-art real-time detectors, including the latest YOLO series (YOLOv10 [33], YOLOv11 [11], YOLOv12 [32], and YOLOv13 [17]) and DETR-based detectors (RT-DETR [38], RT-DETRv2 [23], RT-DETRv3 [36], D-FINE [26], DEIM [15], and DEIMv2 [14]). The results are illustrated in Figure 1, and detailed statistics are provided in Table 1. The results demonstrate that RT-DETRv4 consistently achieves the best performance across all model scales

余弦相似度损失。我们最大化检测器投影特征 F'_5 与教师模型投影特征 F'_7 之间的逐块余弦相似度，其公式化为最小化所有空间位置 (i, j) 上的负余弦相似度平均值：

$$\mathcal{L}_{DSI}(F'_5, F'_7) = -\frac{1}{H_5 W_5} \sum_{i,j} \frac{F'_5(i, j) \cdot F'_7(i, j)}{\|F'_5(i, j)\| \|F'_7(i, j)\|}. \quad (5)$$

3.3. 梯度引导的自适应调制

为确保稳定且自适应的语义监督，我们提出了一种动态梯度引导自适应调制（GAM）机制，该机制根据AIFI模块的*gradient norm ratio*而非原始损失大小来调节其相对贡献。这种基于梯度的调节能够自适应地将AIFI的有效贡献维持在期望范围内，从而实现模型组件间的均衡优化。

具体来说，对于每个训练周期 e 内的训练步骤 t ，我们计算每个主要组件（包括骨干网络、AIFI、CCFF和解码器）的梯度 L_1 范数：

$$\mathcal{C} = \{\text{Backbone, AIFI, CCFF, Decoder}\}, \quad (6)$$

$$G_t^{(C)} = \|\nabla_{\theta_C} \mathcal{L}_{total}\|_1. \quad (7)$$

总梯度幅值由下式给出：

$$G_t^{(total)} = \sum_C G_t^{(C)}, \quad (8)$$

而AIFI在步骤 t 处的相对梯度贡献定义为：

$$r_t = \frac{G_t^{(\text{AIFI})}}{G_t^{(total)}}. \quad (9)$$

随后，我们在第 e 个周期内对所有训练步骤的梯度比进行平均，得到：

$$\bar{r}_e = \frac{1}{T_e} \sum_{t=1}^{T_e} r_t, \quad (10)$$

其中 T_e 表示在时期 e 中的步数。

两个超参数控制调制过程：

- 目标比率（ ρ ）：AIFI的理想平均梯度比率，代表其对优化的理想相对贡献。
- 公差区间（ δ ）：一个定义目标比例周围可接受偏差范围 $[\rho - \delta, \rho + \delta]$ 的裕度。

在每个epoch结束时，GAM会检查 $\bar{r}_e$ 是否处于目标区间内。如果 $\bar{r}_e \in [\rho - \delta, \rho + \delta]$ ，则 $\mathcal{L}_{DSI}$ 的权重 λ_e 保持不变。否则，将调整 λ_e ，使得下一个epoch的AIFI梯度比被导向目标区间。

further boundary 目标范围而非其中点的梯度，因为只有部分AIFI的梯度来源于 $\mathcal{L}_{DSI}$ ，基于边界的调整在平衡点附近能带来更稳定的收敛。

$$\lambda_{e+1} = \begin{cases} \lambda_e \cdot \frac{\rho - \delta}{\bar{r}_e}, & \text{if } \bar{r}_e > \rho + \delta, \\ \lambda_e \cdot \frac{\rho + \delta}{\bar{r}_e}, & \text{if } \bar{r}_e < \rho - \delta, \\ \lambda_e, & \text{otherwise.} \end{cases} \quad (11)$$

此更新规则驱动AIFI的有效梯度贡献在期望的操作范围内收敛，同时防止振荡。超参数 ρ 和 δ 为训练动态提供了显式控制： ρ 定义了期望的监督强度，而 δ 则调节响应性与稳定性之间的权衡。较小的 δ 能实现更快的适应但可能带来不稳定风险，而较大的 δ 则会产生更平稳但更慢的收敛。在实践中，GAM能提供稳定的收敛，并在无需额外调优开销的情况下持续提升语义对齐效果。

4. 实验

4.1. 设置

数据集与评估指标。所有实验均在COCO 2017 [20]数据集上进行，使用train2017划分进行训练，val2017划分进行评估。我们报告标准的COCO评估指标，包括AP（在0.50至0.95范围内以0.05为步长均匀采样的IoU阈值上的平均值）、AP₅₀、AP₇₅，以及不同尺度下的AP：AP_S、AP_M、AP_L。

实现细节。我们的实验基于RT-DETR架构[38]，并融合了RT-DETRv2[23]、D-FINE[26]和DEIM[15]的架构与训练优化。为公平比较，核心超参数与对应基线保持一致。DSI采用预训练且冻结的DINOv3-ViT-B模型作为语义教师。所有评估均使用COCO AP指标，推理延迟（毫秒级）在单张NVIDIA T4 GPU上基于TensorRT FP16精度测得。

4.2. 与SOTA的对比

我们将提出的RT-DETRv4与近期先进的实时检测器进行比较，包括最新的YOLO系列（YOLOv10 [33]、YOLOv11 [11]、YOLOv12 [32]和YOLOv13 [17]）以及基于DETR的检测器（RT-DETR [38]、RT-DETRv2 [23]、RT-DETRv3 [36]、D-FINE [26]、DEIM [15]和DEIMv2 [14]）。结果如图1所示，详细统计数据见表1。实验表明，RT-DETRv4在所有模型规模下均能持续取得最佳性能。

Table 1. **Comparison with other real-time object detectors on COCO [20] val_{2017}** . Results are sourced from the official publications. Values that were not explicitly reported but derived from publicly available weights via standard evaluation are marked with *. R18, R34, R50, and R101 refer to ResNet-18, ResNet-34, ResNet-50, and ResNet-101, respectively.

Model	#Epochs	#Params.	GFLOPs	Latency (ms)	AP^{val}	AP_{50}^{val}	AP_{75}^{val}	AP_S^{val}	AP_M^{val}	AP_L^{val}
YOLOv10-S [33]	500	7	22	2.52	46.3	63.0	50.4	26.8	51.0	63.8
YOLO11-S [11]	500	9	22	2.60	47.0	63.4*	50.5*	-	-	-
YOLOv12-S [32]	600	9	21	2.78	48.0	65.0	51.8	29.8	<u>53.2</u>	65.6
YOLOv13-S [17]	600	9	21	2.98	48.0	65.2	52.0	-	-	-
RT-DETR-R18 [38]	120	20	60	4.61	46.5	63.8	50.4	28.4	49.8	63.0
RT-DETRv2-S [23]	120	20	60	4.61	48.1	65.1	52.1*	30.2*	51.2*	64.2*
RT-DETRv3-R18 [36]	120	20	60	4.61	48.1	65.6	52.0*	30.2*	51.5*	63.9*
D-FINE-S [26]	120	10	25	3.66	48.5	65.6	52.6	29.1	52.2	65.4
DEIM-S [15]	120	10	25	3.66	<u>49.0</u>	<u>65.9</u>	<u>53.1</u>	30.4	52.6	65.7
RT-DETRv4-S (ours)	120	10	25	3.66	49.7	66.8	54.1	<u>30.2</u>	53.6	66.9
YOLOv10-M [33]	500	15	59	4.74	51.1	68.1	55.8	33.8	56.5	67.0
YOLO11-M [11]	500	20	68	4.85	51.5	68.1*	55.8*	-	-	-
YOLOv12-M [32]	600	20	68	4.96	52.5	69.6	57.1	35.7	58.2	68.8
RT-DETR-R34 [38]	120	31	92	6.91	48.9	66.8	52.9	30.6	52.4	66.3
RT-DETRv2-M [23]	120	31	92	6.91	49.9	67.5	54.1*	32.0*	53.2*	66.5*
RT-DETRv3-R34 [36]	120	31	92	6.91	49.9	67.7	53.9*	31.7*	54.0*	66.2*
D-FINE-M [26]	120	19	57	5.91	52.3	69.8	56.4	33.2	56.5	<u>70.2</u>
DEIM-M [15]	90	19	57	5.91	<u>52.7</u>	<u>70.0</u>	<u>57.3</u>	<u>35.3</u>	56.7	69.5
RT-DETRv4-M (ours)	90	19	57	5.91	53.5	71.1	58.1	34.9	<u>57.7</u>	72.1
YOLOv10-L [33]	500	24	120	7.38	53.2	70.1	58.1	35.8	58.5	69.4
YOLO11-L [11]	500	25	87	6.33	53.4	69.7*	58.3*	-	-	-
YOLOv12-L [32]	600	27	89	6.85	53.7	70.7	58.5	<u>36.9</u>	59.5	69.9
YOLOv13-L [17]	600	88	28	8.63	53.4	70.9	58.1	-	-	-
RT-DETR-R50 [38]	72	42	136	9.29	53.1	71.3	57.7	34.8	58.0	70.0
RT-DETRv2-L [23]	72	42	136	9.29	53.4	71.6	57.4*	36.1*	57.9*	70.8*
RT-DETRv3-R50 [36]	120	42	136	9.29	53.4	71.7	57.3*	35.4*	57.4*	69.8*
D-FINE-L [26]	72	31	91	8.07	54.0	71.6	58.4	36.5	58.0	71.9
DEIM-L [15]	50	31	91	8.07	<u>54.7</u>	<u>72.4</u>	<u>59.4</u>	<u>36.9</u>	<u>59.6</u>	71.8
RT-DETRv4-L (ours)	50	31	91	8.07	55.4	73.0	60.3	37.1	60.1	72.9
YOLOv10-X [33]	500	30	160	10.45	54.4	71.3	59.3	37.0	59.8	70.9
YOLO11-X [11]	500	57	195	11.50	54.7	71.3*	59.7*	-	-	-
YOLOv12-X [32]	600	59	199	11.80	55.2	72.0	60.2	39.6	60.7	70.9
YOLOv13-X [17]	600	64	199	14.67	54.8	72.0	59.8	-	-	-
RT-DETR-R101 [38]	72	76	259	13.88	54.3	72.7	58.6	36.0	58.8	72.1
RT-DETRv2-X [23]	72	76	259	13.88	54.3	72.8	58.8*	35.8*	58.8*	72.1*
RT-DETRv3-R101 [36]	120	76	259	13.88	54.6	73.1	-	-	-	-
D-FINE-X [26]	72	62	202	12.90	55.8	73.7	60.2	37.3	60.5	73.4
DEIM-X [15]	50	62	202	12.90	<u>56.5</u>	<u>74.0</u>	<u>61.5</u>	38.8	<u>61.4</u>	<u>74.2</u>
RT-DETRv4-X (ours)	50	62	202	12.90	57.0	74.6	62.1	<u>39.5</u>	61.9	74.8

(S, M, L, and X) without introducing any extra inference and deployment overhead.

Specifically, our **RT-DETRv4-L** achieves **55.4 AP** on COCO at **124 FPS**, outperforming YOLOv13-L (53.4 AP) and DEIM-L (54.7 AP) under comparable or even lower computational budgets. The largest variant, **RT-DETRv4-X**, reaches **57.0 AP**, exceeding DEIM-X (56.5 AP) without introducing any inference overhead. At smaller scales, **RT-DETRv4-S** and **RT-DETRv4-M** obtain **49.7** and **53.5 AP**, respectively, both clearly surpassing their DEIM counter-

parts (49.0 and 52.7 AP).

To ensure a fair comparison within a similar latency regime, we report DEIMv2 [14] results only in Figure 1 and exclude them from Table 1, as their models generally exhibit higher inference latency. Under comparable inference speeds, our RT-DETRv4-M surpasses DEIMv2-S by a large margin (53.5 AP vs. 50.9 AP at 169 FPS vs. 173 FPS), and RT-DETRv4-L further outperforms DEIMv2-M (55.4 AP vs. 53.0 AP at 124 FPS vs. 113 FPS). These results fully demonstrate the effectiveness and great poten-

表1. 在COCO [20] val2017数据集上与其他实时目标检测器的对比。结果来源于官方出版物。未明确报告但通过公开权重经标准评估得出的数值以*标记。R18、R34、R50和R101分别指代ResNet-18、ResNet-34、ResNet-50和ResNet-101。

Model	#Epochs	#Params.	GFLOPs	Latency (ms)	AP^{val}	AP_{50}^{val}	AP_{75}^{val}	AP_S^{val}	AP_M^{val}	AP_L^{val}
YOLOv10-S [33]	500	7	22	2.52	46.3	63.0	50.4	26.8	51.0	63.8
YOLO11-S [11]	500	9	22	2.60	47.0	63.4*	50.5*	-	-	-
YOLOv12-S [32]	600	9	21	2.78	48.0	65.0	51.8	29.8	<u>53.2</u>	65.6
YOLOv13-S [17]	600	9	21	2.98	48.0	65.2	52.0	-	-	-
RT-DETR-R18 [38]	120	20	60	4.61	46.5	63.8	50.4	28.4	49.8	63.0
RT-DETRv2-S [23]	120	20	60	4.61	48.1	65.1	52.1*	30.2*	51.2*	64.2*
RT-DETRv3-R18 [36]	120	20	60	4.61	48.1	65.6	52.0*	30.2*	51.5*	63.9*
D-FINE-S [26]	120	10	25	3.66	48.5	65.6	52.6	29.1	52.2	65.4
DEIM-S [15]	120	10	25	3.66	<u>49.0</u>	<u>65.9</u>	<u>53.1</u>	30.4	52.6	65.7
RT-DETRv4-S (ours)	120	10	25	3.66	49.7	66.8	54.1	<u>30.2</u>	53.6	66.9
YOLOv10-M [33]	500	15	59	4.74	51.1	68.1	55.8	33.8	56.5	67.0
YOLO11-M [11]	500	20	68	4.85	51.5	68.1*	55.8*	-	-	-
YOLOv12-M [32]	600	20	68	4.96	52.5	69.6	57.1	35.7	58.2	68.8
RT-DETR-R34 [38]	120	31	92	6.91	48.9	66.8	52.9	30.6	52.4	66.3
RT-DETRv2-M [23]	120	31	92	6.91	49.9	67.5	54.1*	32.0*	53.2*	66.5*
RT-DETRv3-R34 [36]	120	31	92	6.91	49.9	67.7	53.9*	31.7*	54.0*	66.2*
D-FINE-M [26]	120	19	57	5.91	52.3	69.8	56.4	33.2	56.5	<u>70.2</u>
DEIM-M [15]	90	19	57	5.91	<u>52.7</u>	<u>70.0</u>	<u>57.3</u>	<u>35.3</u>	56.7	69.5
RT-DETRv4-M (ours)	90	19	57	5.91	53.5	71.1	58.1	34.9	<u>57.7</u>	72.1
YOLOv10-L [33]	500	24	120	7.38	53.2	70.1	58.1	35.8	58.5	69.4
YOLO11-L [11]	500	25	87	6.33	53.4	69.7*	58.3*	-	-	-
YOLOv12-L [32]	600	27	89	6.85	53.7	70.7	58.5	<u>36.9</u>	59.5	69.9
YOLOv13-L [17]	600	88	28	8.63	53.4	70.9	58.1	-	-	-
RT-DETR-R50 [38]	72	42	136	9.29	53.1	71.3	57.7	34.8	58.0	70.0
RT-DETRv2-L [23]	72	42	136	9.29	53.4	71.6	57.4*	36.1*	57.9*	70.8*
RT-DETRv3-R50 [36]	120	42	136	9.29	53.4	71.7	57.3*	35.4*	57.4*	69.8*
D-FINE-L [26]	72	31	91	8.07	54.0	71.6	58.4	36.5	58.0	71.9
DEIM-L [15]	50	31	91	8.07	<u>54.7</u>	<u>72.4</u>	<u>59.4</u>	<u>36.9</u>	<u>59.6</u>	71.8
RT-DETRv4-L (ours)	50	31	91	8.07	55.4	73.0	60.3	37.1	60.1	72.9
YOLOv10-X [33]	500	30	160	10.45	54.4	71.3	59.3	37.0	59.8	70.9
YOLO11-X [11]	500	57	195	11.50	54.7	71.3*	59.7*	-	-	-
YOLOv12-X [32]	600	59	199	11.80	55.2	72.0	60.2	39.6	60.7	70.9
YOLOv13-X [17]	600	64	199	14.67	54.8	72.0	59.8	-	-	-
RT-DETR-R101 [38]	72	76	259	13.88	54.3	72.7	58.6	36.0	58.8	72.1
RT-DETRv2-X [23]	72	76	259	13.88	54.3	72.8	58.8*	35.8*	58.8*	72.1*
RT-DETRv3-R101 [36]	120	76	259	13.88	54.6	73.1	-	-	-	-
D-FINE-X [26]	72	62	202	12.90	55.8	73.7	60.2	37.3	60.5	73.4
DEIM-X [15]	50	62	202	12.90	<u>56.5</u>	<u>74.0</u>	<u>61.5</u>	38.8	<u>61.4</u>	<u>74.2</u>
RT-DETRv4-X (ours)	50	62	202	12.90	57.0	74.6	62.1	<u>39.5</u>	61.9	74.8

(S、M、L 和 X)，无需引入任何额外的推理和部署开销。

具体而言，我们的RT-DETRv4-L在COCO数据集上以124 FPS的速度实现了55.4 AP，在计算量相当甚至更低的条件下，超越了YOLOv13-L (53.4 AP) 和DEIM-L (54.7 AP)。最大规模的变体RT-DETRv4-X达到了57.0 AP，在未引入任何推理开销的情况下超越了DEIM-X (56.5 AP)。在较小规模上，RT-DETRv4-S和RT-DETRv4-M分别取得了49.7和53.5 AP，均明显优于同规模的DEIM对应模型。

部分 (49.0 和 52.7 AP)。

为确保在相似的延迟范围内进行公平比较，我们仅在图表1中报告DEIMv2 [14]的结果，并将其排除在表格1之外，因为它们的模型通常表现出更高的推理延迟。在可比较的推理速度下，我们的RT-DETRv4-M大幅超越DEIMv2-S (169 FPS对比173 FPS时，AP值为53.5对比50.9)，而RT-DETRv4-L进一步优于DEIMv2-M (124 FPS对比113 FPS时，AP值为55.4对比53.0)。这些结果充分证明了其有效性和巨大潜力。

Table 2. **Results of ablation on DSI and GAM across multiple detectors.** Our method brings consistent and significant gains with zero additional inference cost.

Method	AP	AP ₅₀	AP ₇₅
RT-DETRv2-L	52.1	70.2	56.7
w/ DSI	52.3 (+0.2)	70.4 (+0.2)	56.4 (-0.3)
w/ DSI+GAM	52.6 (+0.5)	70.7 (+0.5)	56.8 (+0.1)
D-FINE-L	53.1	70.8	57.4
w/ DSI	53.2 (+0.1)	70.8 (+0)	57.7 (+0.3)
w/ DSI+GAM	53.4 (+0.3)	71.1 (+0.3)	58.0 (+0.6)
DEIM-L	53.8	71.4	58.5
w/ DSI	53.9 (+0.1)	71.3 (-0.1)	58.8 (+0.3)
w/ DSI+GAM	54.3 (+0.5)	71.8 (+0.4)	59.0 (+0.5)

tial of the proposed method. Although DEIMv2-X achieves stronger performance, its latency is also higher than that of RT-DETRv4-X. Moreover, directly adopting DINOv3 as the backbone to obtain semantic richness is fundamentally constrained by model size and deployment cost, limiting it to the Tiny or Small variants of DINOv3 and making it difficult to scale to more powerful large-scale models. In contrast, our framework remains agnostic to both VFM type and scale, introducing no inference or deployment overhead, offering a more flexible and deployment-friendly solution.

4.3. Ablation Study

We conduct a series of ablation experiments to verify the effectiveness of proposed modules. Unless stated otherwise, all ablation experiments are trained for **36 epochs**. Unspecified hyperparameters or configurations follow the best settings for the corresponding experimental setup.

Ablation on DSI and GAM. We first assess the effectiveness of DSI and GAM. As shown in Table 2, applying DSI can bring a slight performance improvement, while further applying GAM can significantly improve the performance gain (0.5 AP), which fully proves the effectiveness of both. To verify the general applicability of our method, we also conduct experiments on RT-DETRv2 and D-FINE, and the results show that our method can bring consistent performance gains to different detectors.

Ablation on semantic injection position. To validate the choice of injection position, we compare the strategies shown in Figure 3. Results in Table 3 indicate that directly applying semantic supervision to backbone features (S_3 , S_4 , or S_5) individually or jointly yields no improvement. Similarly, the hybrid approach (strategy (b)) that aligns both backbone and F_5 features provides no gain (53.8 AP).

In contrast, our design (strategy (c)), which aligns only the AIFI output F_5 , achieves a clear 0.5 AP improvement (54.3 AP). This demonstrates that maintaining richer semantics in F_5 is crucial for enhancing detection performance, as it plays a key role in propagating high-level semantic infor-

Table 3. **Results of ablation on semantic injection position.** We compare the effectiveness of applying DSI at different position, corresponding to the strategies in Figure 3. Aligning only the AIFI output (F_5) yields the best performance.

Position	S_3	S_4	S_5	F_5	AP	AP ₅₀	AP ₇₅
Baseline	-	-	-	-	53.8	71.4	58.5
Backbone	✓	-	-	-	53.7	71.2	58.4
	-	✓	-	-	53.7	71.3	58.4
	-	-	✓	-	53.8	71.3	58.5
	✓	✓	✓	-	53.7	71.3	58.5
Hybrid	✓	✓	✓	✓	53.8	71.4	58.4
AIFI	-	-	-	✓	54.3	71.8	59.0

Table 4. **Results of ablation on the feature projector.**

Projector Arch.	AP	AP ₅₀	AP ₇₅
Baseline	53.8	71.4	58.5
w/ 1x1 Conv	53.8	71.5	58.5
w/ MLP	54.2	71.7	59.0
w/ Linear	54.3 (+0.5)	71.8 (+0.4)	59.0 (+0.5)

Table 5. **Results of ablation on the alignment loss.** The cosine similarity loss demonstrates superior performance. DEIM-M is adopted as the baseline. All reported results are obtained from models trained for 90 epochs with 12 epochs EMA following the training protocol of DEIM.

Loss Function	AP	AP ₅₀	AP ₇₅
Baseline	52.5	69.9	57.2
w/ MSE Loss	52.7	70.0	57.4
w/ Cosine Similarity	53.5 (+1.0)	71.1 (+1.2)	58.1 (+0.9)

mation to subsequent feature hierarchies. Furthermore, the ineffectiveness of the hybrid approach suggests that simultaneously aligning features from the CNN-based backbone and the Transformer-based AIFI may introduce optimization conflicts or semantic misalignment. Our chosen strategy is not only more effective but also more efficient. It avoids the complexity of multiple intermediate projections and interpolations, and the gradient from the single F_5 alignment loss naturally flows back to synergistically update both AIFI and the backbone, achieving consistent enhancement with a single, targeted objective.

Design of the feature projector. The feature projector is a crucial component for bridging the student and teacher feature spaces. We explore several architectural choices, as detailed in Table 4. A linear-based projector yields the best results, striking an optimal balance between expressive power and parameter efficiency.

Choice of loss function. We further study the alignment loss of DSI. We compare the Mean Squared Error (MSE) loss and the Cosine Similarity loss in Table 5, with the latter

表2. 在多种检测器上对DSI和GAM的消融实验结果。我们的方法以零额外推理成本带来了持续且显著的性能提升。

Method	AP	AP ₅₀	AP ₇₅
RT-DETRv2-L	52.1	70.2	56.7
w/ DSI	52.3 (+0.2)	70.4 (+0.2)	56.4 (-0.3)
w/ DSI+GAM	52.6 (+0.5)	70.7 (+0.5)	56.8 (+0.1)
D-FINE-L	53.1	70.8	57.4
w/ DSI	53.2 (+0.1)	70.8 (+0)	57.7 (+0.3)
w/ DSI+GAM	53.4 (+0.3)	71.1 (+0.3)	58.0 (+0.6)
DEIM-L	53.8	71.4	58.5
w/ DSI	53.9 (+0.1)	71.3 (-0.1)	58.8 (+0.3)
w/ DSI+GAM	54.3 (+0.5)	71.8 (+0.4)	59.0 (+0.5)

所提方法的潜力。尽管DEIMv2-X实现了更强的性能，但其延迟也高于RT-DETRv4-X。此外，直接采用DINO v3作为骨干网络以获取语义丰富性，从根本上受到模型规模和部署成本的限制，只能局限于DINOv3的Tiny或Small变体，难以扩展到更强大的大规模模型。相比之下，我们的框架对视觉基础模型（VFM）的类型和规模均保持不可知性，不引入任何推理或部署开销，提供了一个更灵活且更利于部署的解决方案。

4.3. 消融研究

我们进行了一系列消融实验以验证所提出模块的有效性。除非另有说明，所有消融实验均训练36个周期。未明确指定的超参数或配置遵循相应实验设置中的最佳设定。

关于DSI和GAM的消融实验。我们首先评估了DSI和GAM的有效性。如表2所示，应用DSI能带来轻微的性能提升，而进一步应用GAM则能显著提高性能增益（0.5 AP），这充分证明了两者的有效性。为验证我们方法的普适性，我们还在RT-DETRv2和D-FINE上进行了实验，结果表明我们的方法能为不同检测器带来持续的性能提升。

语义注入位置的消融实验。为了验证注入位置的选择，我们比较了图3所示的策略。表3的结果表明，直接将语义监督应用于骨干特征（ S_3 、 S_4 或 S_5 ）单独或联合使用均未带来改进。同样，对齐骨干特征与 F_5 特征的混合方法（策略(b)）也未带来增益（53.8 AP）。

相比之下，我们的设计（策略(c)）仅对齐AIFI输出 F_5 ，实现了明显的0.5 AP提升（54.3 AP）。这表明在 F_5 中保持更丰富的语义对于提升检测性能至关重要，因为它在传播高层语义信息中起着关键作用。

表3. 语义注入位置消融实验结果。我们比较了在不同位置应用DSI的效果，对应图3中的策略。仅对齐AIFI输出($\{v^*\}$)可获得最佳性能。

Position	S_3	S_4	S_5	F_5	AP	AP ₅₀	AP ₇₅
Baseline	-	-	-	-	53.8	71.4	58.5
Backbone	✓	-	-	-	53.7	71.2	58.4
	-	✓	-	-	53.7	71.3	58.4
	-	-	✓	-	53.8	71.3	58.5
	✓	✓	✓	-	53.7	71.3	58.5
Hybrid	✓	✓	✓	✓	53.8	71.4	58.4
AIFI	-	-	-	✓	54.3	71.8	59.0

表4. 特征投影器消融实验结果。

Projector Arch.	AP	AP ₅₀	AP ₇₅
Baseline	53.8	71.4	58.5
w/ 1x1 Conv	53.8	71.5	58.5
w/ MLP	54.2	71.7	59.0
w/ Linear	54.3 (+0.5)	71.8 (+0.4)	59.0 (+0.5)

表5. 对齐损失消融实验结果。余弦相似度损失展现出更优性能。实验以DEIM-M作为基线模型，所有报告结果均遵循DEIM训练协议，采用90轮训练周期及12轮指数移动平均策略获得。

Loss Function	AP	AP ₅₀	AP ₇₅
Baseline	52.5	69.9	57.2
w/ MSE Loss	52.7	70.0	57.4
w/ Cosine Similarity	53.5 (+1.0)	71.1 (+1.2)	58.1 (+0.9)

信息传递至后续的特征层次。此外，混合方法的有效性表明，同时对齐基于CNN的主干网络和基于Transformer的AIFI的特征可能会引入优化冲突或语义不对齐。我们选择的策略不仅更有效，而且更高效。它避免了多个中间投影和插值的复杂性，并且来自单一 $\{v^*\}$ 对齐损失的梯度自然回流，协同更新AIFI和主干网络，通过单一、有针对性的目标实现一致的增强。

特征投影器的设计。特征投影器是连接学生和教师特征空间的关键组件。我们探索了多种架构选择，详见表4。基于线性的投影器取得了最佳效果，在表达能力和参数效率之间实现了最优平衡。

损失函数的选择。我们进一步研究了DSI的对齐损失。我们在表5中比较了均方误差（MSE）损失和余弦相似度损失，后者

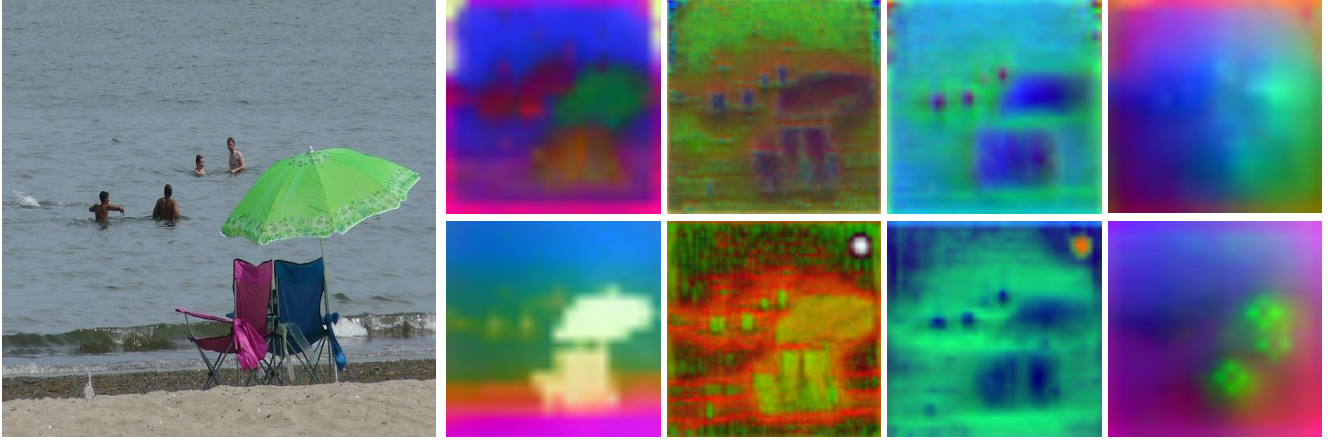


Figure 4. **Comparison of dense features.** We compare the feature map quality of DEIM-L (top) and RT-DETRv4-L (bottom) by projecting dense outputs to RGB space using PCA. The visualization reveals that our DSI module substantially enhances the semantic representation of AIFI features, which in turn benefits subsequent CCFF features. From left to right: input image, AIFI feature map F_5 , and multi-scale CCFF features P_3 , P_4 , P_5 .

Table 6. **Ablation on the loss weighting strategy.** GAM consistently surpasses the best-tuned static weight. DEIM-L is adopted as the baseline. All reported results are obtained from models trained for 50 epochs with 8 epochs EMA following [15].

λ	0.1	0.2	0.4	1	2	4
AP^{val}	54.7	54.7	54.8	54.9	<u>55.0</u>	<u>55.0</u>
λ	<u>10</u>	20	<u>30</u>	50	100	GAM
AP^{val}	<u>55.0</u>	55.1	<u>55.0</u>	54.9	54.6	55.4

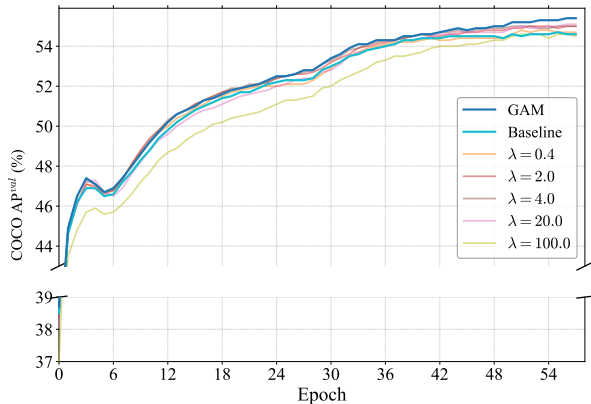


Figure 5. **Validation AP evolution on COCO during training.** We compare our dynamic GAM with a baseline model and several static λ values for the DSI loss. The GAM strategy consistently outperforms all static configurations, showcasing its ability to provide stable and effective supervision throughout the training process.

outperforming the former, validating our hypothesis that aligning feature direction is more crucial.

Comparison of GAM and static weights. Finally, we validate the proposed GAM against static weights. The su-

periority of GAM in navigating the training dynamics is further illustrated in Figure 5, which plots the validation AP over epochs. As shown, the curve for GAM consistently remains above the baseline and all static weight configurations, demonstrating a clear and stable performance advantage throughout the training process.

Table 6 details the results for static weight and GAM. For static weighting, performance peaks at $\lambda = 20$, achieving 55.1 AP, but degrades with either smaller or larger values. However, observing the training curve in Figure 5, we find that even this optimal static weight ($\lambda = 20$) leads to slower convergence in the early-to-mid stages, highlighting the inherent limitations of a fixed hyperparameter. Our experiments indicate that GAM achieves the best performance (55.4 AP).

Feature visualization. Figure 4 visually compares the feature maps between DEIM-L and our RT-DETRv4-L. Notably, our model enriches the semantic content of the AIFI feature F_5 , leading to more precise and distinguishable object contours and backgrounds across the subsequent multi-scale features P_3 , P_4 , and P_5 . In particular, P_5 exhibits a markedly stronger and more concentrated response to object regions.

5. Discussion

To further highlight the advantages of our framework over existing methods [14, 26], we discuss it from three perspectives: deployment efficiency, scalability, and training efficiency.

Deployment Efficiency. Our method introduces zero modification to detector architectures and does not alter the inference pipeline, ensuring that no additional computational cost or latency is incurred. This deployment-friendly property is crucial for industrial applications, where real-time detectors are tightly integrated into existing systems and hardware-constrained environments.

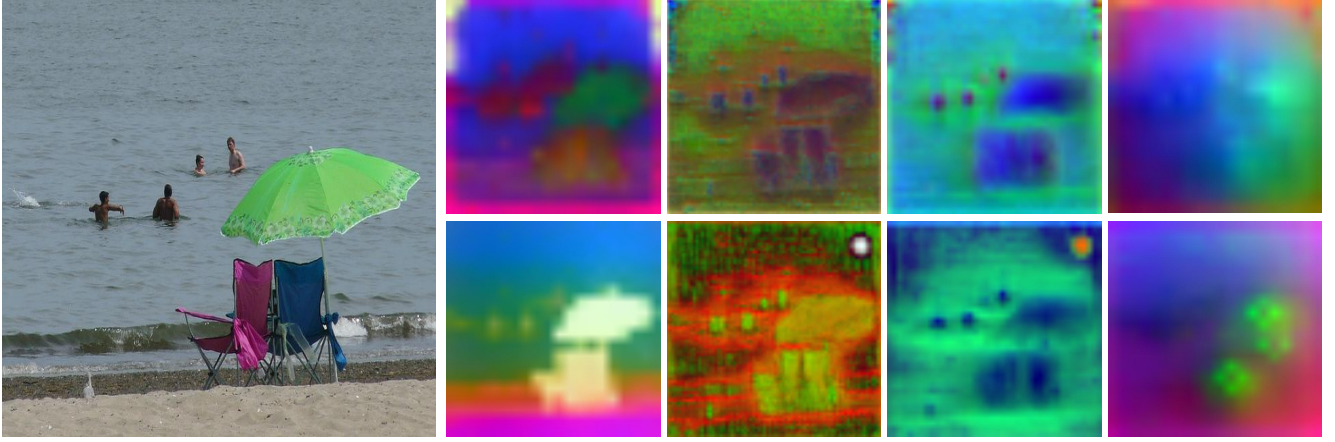


图4. 密集特征对比。我们通过PCA将密集输出投影至RGB空间，对比了DEIM-L（上）与RT-DETRv4-L（下）的特征图质量。可视化结果表明，我们的DSI模块显著增强了AIFI特征的语义表示，进而提升了后续CCFF特征的性能。从左至右分别为：输入图像、AIFI特征图 F_5 以及多尺度CCFF特征 P_3 、 P_4 、 P_5 。

表6. 损失加权策略的消融研究。GAM持续超越最佳调优的静态权重。采用DEIM-L作为基线。所有报告结果均来自按照[15]方法训练50个周期并采用8周期指数移动平均的模型。

λ	0.1	0.2	0.4	1	2	4
AP^{val}	54.7	54.7	54.8	54.9	<u>55.0</u>	<u>55.0</u>
λ	10	20	30	50	100	GAM
AP^{val}	<u>55.0</u>	55.1	<u>55.0</u>	54.9	54.6	55.4

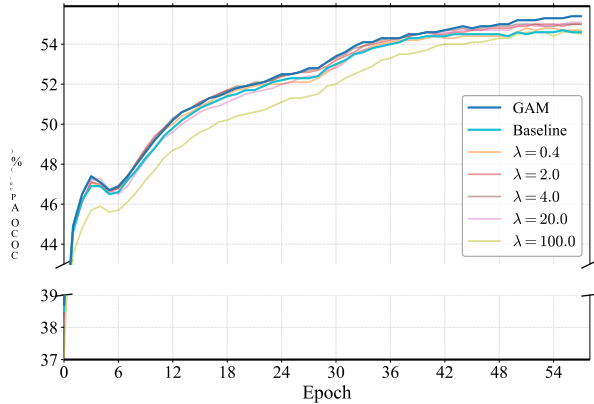


图5. 训练期间COCO数据集上的验证AP变化。我们将动态GAM与基线模型及DSI损失的多个静态 λ 值进行比较。GAM策略始终优于所有静态配置，展现了其在训练全程提供稳定有效监督的能力。

优于前者，验证了我们的假设，即对齐特征方向更为关键。

GAM与静态权重的比较。最后，我们通过静态权重验证所提出的GAM。该

GAM在引导训练动态方面的优越性在图5中得到了进一步说明，该图绘制了验证AP随训练轮次的变化曲线。如图所示，GAM的曲线始终保持在基线及所有静态权重配置之上，在整个训练过程中展现出清晰而稳定的性能优势。

表6详细列出了静态权重和GAM的结果。对于静态加权，性能在 $\lambda = 20$ 时达到峰值，获得55.1 AP，但数值过小或过大时性能均会下降。然而通过观察图5中的训练曲线，我们发现即使这种最优静态权重（ $\lambda = 20$ ）也会导致训练中前期收敛速度较慢，这凸显了固定超参数的内在局限性。我们的实验表明，GAM实现了最佳性能（55.4 AP）。

特征可视化。图4直观比较了DEIM-L与我们的RT-DETRv4-L的特征图。值得注意的是，我们的模型丰富了AIFI特征 F_5 的语义内容，使得后续多尺度特征 P_3 、 P_4 和 P_5 中的物体轮廓与背景更为精确且可区分。特别是 P_5 对物体区域表现出显著更强且更集中的响应。

5. 讨论

为了进一步凸显我们的框架相较于现有方法[14, 26]的优势，我们从三个角度进行讨论：部署效率、可扩展性和训练效率。

部署效率。我们的方法对检测器架构进行了零修改，且不改变推理流程，确保不会产生额外的计算成本或延迟。这一部署友好的特性对于工业应用至关重要，因为实时检测器通常紧密集成于现有系统及硬件受限的环境中。

Scalability. The framework is highly general and can be seamlessly applied to detectors with diverse architectures, including CNN-based and transformer-based detectors. It enables all types of real-time detectors to quickly benefit from the rapid progress of Vision Foundation Models (VFM). Moreover, the framework is not restricted to any specific type or scale of VFM. It can flexibly incorporate different foundation models, such as DINOv3 [31], MAE [13], or CLIP [27], and even benefit from arbitrarily large models for distilling semantics into real-time detectors. This flexibility also opens up promising directions for multi-VFM semantic integration, further demonstrating the framework’s generality and scalability.

Training Efficiency. Our approach is lightweight and easy to implement. Since neither the detector structure nor the optimization pipeline is modified by the incorporation of VFMs, the additional training cost remains minimal. This efficiency highlights the practicality of our method for large-scale applications and real-time industrial deployment.

6. Conclusion

In this work, we present a cost-effective and adaptable semantic distillation framework that enhances real-time DETR-based object detectors without increasing inference or deployment overhead. Through the proposed Deep Semantic Injector (DSI) and Gradient-guided Adaptive Modulation (GAM), our method effectively transfers high-level semantics from Vision Foundation Models to lightweight detectors in a stable and task-aligned manner. Extensive experiments on COCO demonstrate consistent and significant performance gains across multiple model scales, culminating in the state-of-the-art RT-DETRv4 series. These results highlight the effectiveness of explicit semantic supervision in bridging the gap between large-scale foundation models and resource-efficient detection architectures. Overall, our work provides a practical pathway toward unlocking the potential of foundation models for efficient visual perception.

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可扩展性。该框架具有高度通用性，可无缝应用于多种架构的检测器，包括基于CNN和基于Transformer的检测器。它使各类实时检测器都能快速受益于视觉基础模型（VFM）的快速发展。此外，该框架不受限于任何特定类型或规模的VFM，能够灵活整合不同的基础模型，例如DINOv3 [31]、MAE [13]或CLIP [27]，甚至可利用任意大规模模型将语义知识蒸馏到实时检测器中。这种灵活性也为多VFM语义融合开辟了前景广阔的研究方向，进一步体现了该框架的通用性与可扩展性。

训练效率。我们的方法轻量且易于实现。由于检测器结构或优化流程均未因融入VFMs而改变，因此额外的训练成本保持在最低水平。这种高效性凸显了我们方法在大规模应用和实时工业部署中的实用性。

6. 结论

在本工作中，我们提出了一种成本效益高且适应性强的语义蒸馏框架，该框架在不增加推理或部署开销的情况下，增强了基于实时DETR的目标检测器。通过提出的深度语义注入器（DSI）和梯度引导自适应调制（GAM），我们的方法以稳定且任务对齐的方式，有效地将视觉基础模型中的高级语义迁移到轻量级检测器中。在COCO数据集上的大量实验表明，该方法在多个模型尺度上均实现了持续且显著的性能提升，最终形成了最先进的RT-DETRv4系列。这些结果突显了显式语义监督在弥合大规模基础模型与资源高效检测架构之间差距方面的有效性。总体而言，我们的工作为释放基础模型在高效视觉感知方面的潜力提供了一条实用途径。

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